CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 6: Simultaneous Linear Equations

Every formula and result of the chapter on one sheet

1	General form	$a_1x + b_1y + c_1 = 0,\ a_2x + b_2y + c_2 = 0$
2	Substitution	Express one variable from one equation; substitute in the other
3	Elimination	Make coefficients equal; add or subtract
4	Cross-multiplication	$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{c_1a_2 - c_2a_1} = \frac{1}{a_1b_2 - a_2b_1}$
5	Unique solution	$a_1b_2 - a_2b_1 \neq 0$
6	Reciprocal form	Put $\frac{1}{x} = a,\ \frac{1}{y} = b$
7	Two-digit number	$10x + y$; reversed $10y + x$
8	Downstream speed	Boat + stream
9	Upstream speed	Boat − stream
10	Distance	Speed × time
11	Symmetric pair	Add and subtract the equations to get $x + y$ and $x - y$